Surface-Dust Recovery and Intermittent Storms in a Seasonal Mars Model

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1 Introduction

Mars has a dusty season every year near perihelion, and every observed planet-encircling (global) dust storm has begun in that season. Yet global storms occur in only about a quarter of Mars years, at irregular intervals [1]. One proposed explanation is that a large storm depletes the loose dust where lifting is easiest, and that dust must be replenished before another can grow [1], [2]. This report asks whether a small model can turn a regular dusty season into intermittent large events. The model combines dust-lifting feedback, a reservoir of liftable surface dust that events deplete and that slowly replenishes, and seasonal forcing that varies modestly between years. Simulations and a fixed-dust equilibrium analysis test this question.

2 Mathematical model

The model is spatially lumped, with no geography and two state variables. Atmospheric dust A is the dust optical depth, a dimensionless measure of how strongly airborne dust blocks sunlight. Available surface dust S is the fraction of liftable dust remaining in the source-region reservoir, from 0 (empty) to 1 (full), not all dust on the surface. Time t is in sols (Martian days), with $P = 669$ sols per Mars year:

$$\frac{dA}{dt} = \frac{A_b}{T_s} + L(A,S,F) - \frac{A}{T_s}, \qquad \frac{dS}{dt} = \frac{1-S}{P\,T_r} - \lambda\,L(A,S,F). \tag{2.1}$$

In the atmospheric equation, dust devils and small storms supply background lifting A_b/T_s, which alone holds A at the typical background optical depth $A_b = 0.5$; storm lifting L adds more; and settling removes dust with timescale $T_s = 43$ sols, measured at the Curiosity rover site after the 2018 global storm and applied here planet-wide [1]. In the surface equation, the reservoir replenishes toward $S = 1$ through a prescribed recovery term with replenishment time T_r Mars years ($P\,T_r$ sols), and storm lifting depletes it at the rate λL. The same lifting term therefore raises A and lowers S; λ converts lifted optical depth into reservoir fraction, and settling does not directly refill the reservoir. Lifting requires both available surface dust and wind:

$$L = k\,S\,\Theta(W), \qquad \Theta(W) = \frac{1}{2}\left[1 + \tanh\left(\frac{W - W_t}{\delta}\right)\right], \qquad W = W_0\,F\,(1+\alpha A)\,e^{-A/A_c}. \tag{2.2}$$

The switch Θ rises smoothly from nearly 0 to 1 as the relative wind strength W passes the lifting threshold W_t, but is not exactly zero below it; W is a dimensionless function of A and the seasonal forcing F, not a wind speed or third equation. Its factor $1 + \alpha A$ represents dust-heating feedback, in which dusty air absorbs sunlight and drives stronger winds; e^{-A/A_c} represents

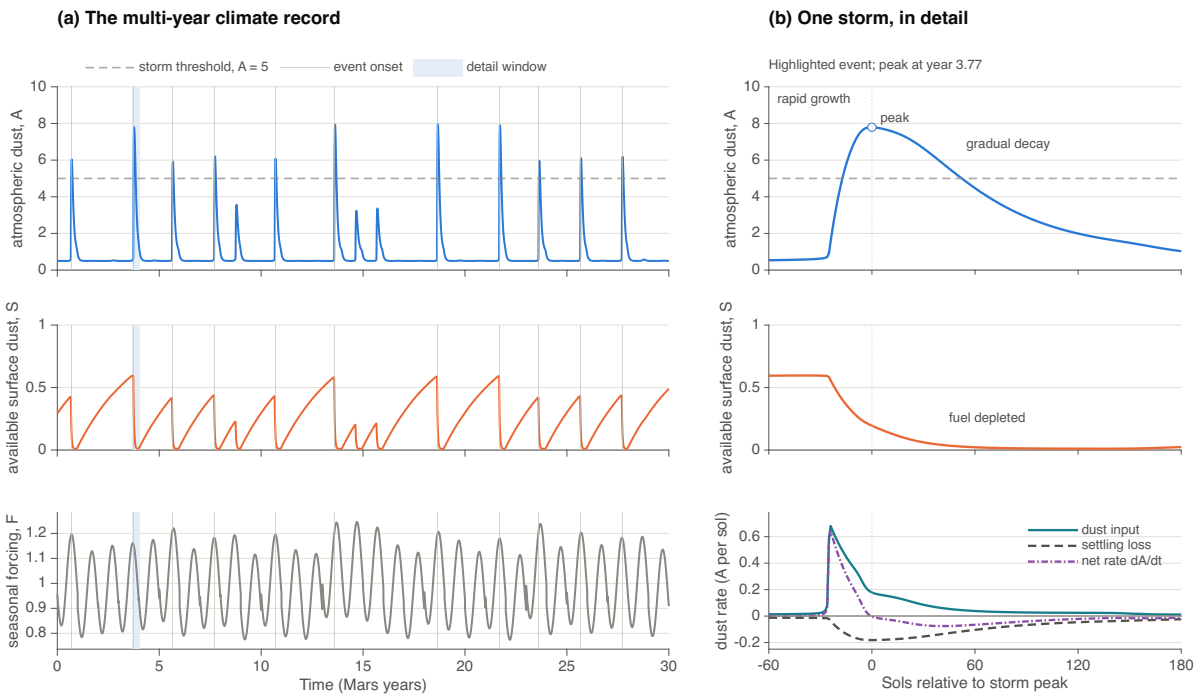


Figure 1. Variable-seasons simulation (annual forcing factors 0.94–1.06, seed 42). (a) A, S, and F over 30 simulated Mars years after a 20-year warm-up, with the $A = 5$ cutoff and the 11 global-event onsets. (b) The shaded window: 60 sols before to 180 sols after the peak of the second detected event. Rates, in A per sol: dust input $A_b/T_s + L$, settling loss $-A/T_s$, and their sum dA/dt.

shading, which weakens surface winds in very dusty air. The seasonal forcing is sunlight relative to its annual mean,

$$F(t) = \left[1 + \varepsilon_F \cos\left(2\pi\left(\frac{t}{P} - \phi_p\right)\right)\right](1 + \eta_n), \tag{2.3}$$

where $\varepsilon_F = 0.18$ makes perihelion forcing 44% stronger than aphelion forcing, close to the observed 40% difference in solar flux [1], and $\phi_p = 0.70$ sets perihelion's timing. With identical seasons, $\eta_n = 0$; with variable seasons, each model year n draws one η_n uniformly from $[-0.06, 0.06]$, scaling that year's curve. The other baseline values are $k = 1.2$ per sol, $\lambda = 0.04$, $T_r = 3$ Mars years, $W_0 = 0.44$, $W_t = 1$, $\delta = 0.05$, $\alpha = 1.7$, and $A_c = 4$.

3 Results

All simulations use MATLAB's stiff solver `ode15s` [3] with relative and absolute tolerances of 10^{-8} and 10^{-10}. Each model year is solved separately, so changes in η_n occur only at solver restarts, and output is saved every sol. Runs start at $A = 0.5$, $S = 0.9$ and discard 20 warm-up years. A model global event begins when A rises through the chosen cutoff of 5 after falling below 2 since the last event; observed global storms can reach optical depths of 5 or more [1].

Figure 1 shows 30 years of a variable-seasons simulation, with both state variables evolving. Although every year has a high-forcing season, the record contains only 11 global events, 1.9 to

5.0 years apart, and three smaller events that stay below $A = 5$. Each global event depletes S to nearly zero, and S then replenishes over the following years, so the reservoir links each event to the next.

The close-up in Figure 1b, chosen for illustration rather than as a typical case, shows the rate balance within one event. Once dust-heating feedback turns the switch on, dust input far exceeds settling and A rises from 1 to 5 in about a week. The same lifting depletes S from about 0.6 before the event to 0.2 at the peak. By then shading has also partly closed the switch, so input falls below settling and dA/dt changes sign. After the peak, the wind strengthens again as A falls, but S keeps declining toward 0.01, so input stays below settling and A decays, at first more slowly than settling alone would allow.

Because A changes within days to weeks while S replenishes over years, it is useful to examine the atmospheric equation alone. Figure 2 evaluates the same net-rate expression as Figure 1b, but holds S and F fixed and varies A. This analysis does not assume that S stays constant during an event; in the evolving storm of Figure 1b, S falls rapidly. Zeros of this rate are equilibria: a zero where the rate changes from positive to negative as A increases is stable, and one where it changes from negative to positive is unstable [4].

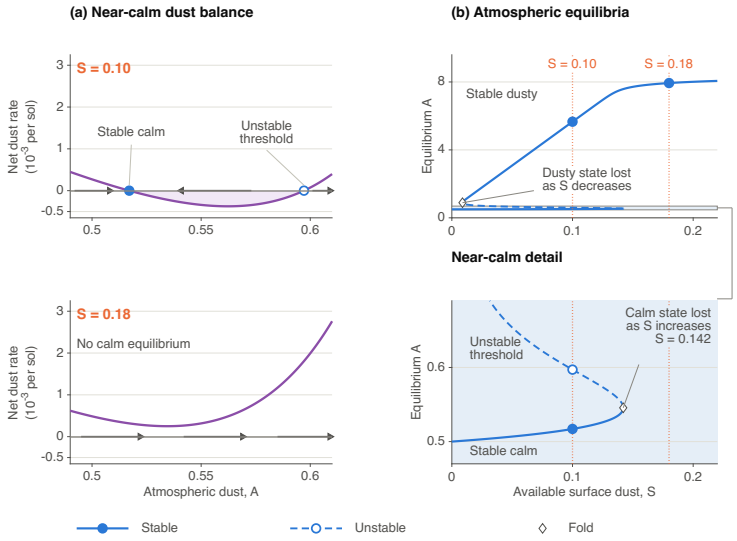


Figure 2. Analysis of the atmospheric equation in (2.1), with forcing fixed at the baseline perihelion value $F = 1.18$ and S fixed in each calculation; it is not a simulated trajectory. (a) Net rate against A for $S = 0.10$ and 0.18; arrows give the direction of change. (b) Equilibrium A against S, with orange guides at the two examples. Solid lines and filled circles are stable, and dashed lines and open circles unstable, to changes in A alone; diamonds mark folds. The lower plot enlarges the outlined near-calm region.

At $S = 0.10$ the near-calm rate has two zeros: a stable calm state at $A \approx 0.52$ and an unstable threshold at $A \approx 0.60$. Small disturbances return to calm, while A pushed past the threshold rises to a third, stable equilibrium near 5.7. At $S = 0.18$ the rate is positive throughout the near-calm range, so the calm state and threshold no longer exist, and A starting there rises toward the only equilibrium, near 7.9.

Figure 2b follows the equilibria as S changes. At a fold, two equilibria merge and disappear, and the rate touches zero without changing sign: the calm state and threshold merge at $S \approx 0.142$, and the dusty state and threshold at $S \approx 0.009$. More available surface dust therefore changes which atmospheric states exist, not only how large an event becomes. Because S falls during an

event and F changes through the year, however, $S \approx 0.142$ is not an exact onset threshold, and losing the calm state does not guarantee a global event within one season.

Figure 2 varies the amount of available surface dust, S; Figure 3 varies how quickly the reservoir replenishes, through T_r. Because the equilibrium diagram cannot say when a seasonal trajectory reaches a global event, this experiment returns to the full model, with both variables and the forcing evolving. Each run reports the share of years whose maximum A reaches 2 or 5. This counts years above each threshold, not separate events, since an episode above $A = 2$ can continue into the next year.

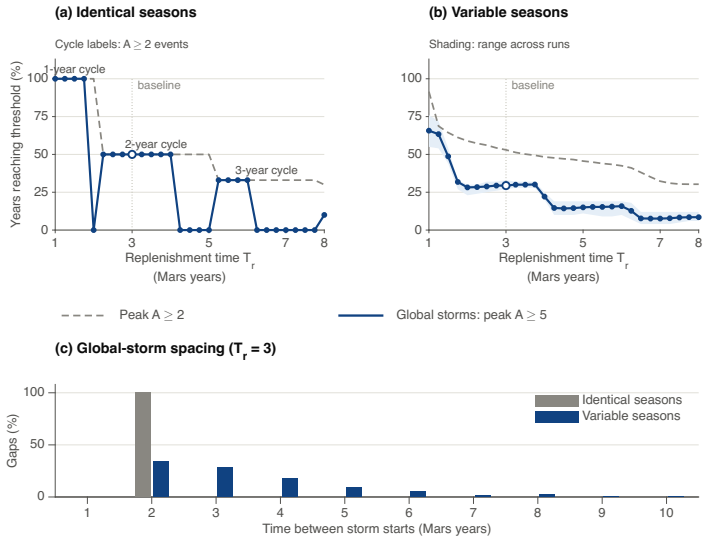


Figure 3. Full-model sweep of T_r (1 to 8 Mars years, step 0.25), other parameters at baseline; each run records 100 Mars years after a 20-year warm-up. (a) Identical seasons, one run per T_r. (b) Variable seasons, ten runs per T_r, reusing seeds 42–51 at every T_r; the band spans the lowest to highest run, not a confidence interval. (c) Intervals between global-event onsets at $T_r = 3$, computed within each run, excluding incomplete waits at the record ends, then pooled (49 identical-season and 284 variable-season intervals); each distribution sums to 100%, in one-year bins centered on integers.

With identical seasons (Figure 3a), events reaching $A \geq 2$ recur on regular one-, two-, and three-year schedules as T_r increases to 7.75 (checked against onset intervals). Global-event frequency, however, does not fall steadily. Within each schedule, slower replenishment leaves less available surface dust when each event begins, so peaks decline until events keep their timing but no longer reach $A = 5$. With variable seasons (Figure 3b), these steps are smoothed. At the baseline $T_r = 3$, global events occur in 50% of years with identical seasons and 29.4% with variable seasons (27–33% across ten 100-year runs); the 11 events in Figure 1 come from a shorter, 30-year portion of the seed-42 run.

Annual occurrence does not describe how events are spaced, so Figure 3c compares intervals between global-event onsets at the same baseline. With identical seasons, every interval is exactly two years. With variable seasons, intervals range from about 1.9 to 10 years, with a mean of 3.4 years; two-year intervals are most common, but three- to five-year intervals are frequent. Because S carries over between years, one season's outcome also shifts later events.

4 Discussion

Taken together, the experiments show how the model's parts interact. Dust-heating feedback creates a threshold for storm growth, and enough available surface dust removes that threshold.

Each global event depletes the reservoir that supplies its lifting, and slow replenishment sets when a later season can produce another event. In the baseline comparison, variable forcing turns the resulting two-year schedule into irregular spacing, much as a low-order circulation model produced year-to-year variability in global storms only once random forcing was added [5].

These results describe the model, not Mars. The seasonal comparison covers one baseline parameter set, so it cannot show whether randomness is required for irregular spacing elsewhere, and no model component was removed to test whether it was necessary. Without geography, the model identifies "global" events by dust level rather than spatial extent. Most parameters are illustrative rather than fitted, so the 29.4% baseline rate is not a test against the observed rate of about 25%. The simulated records are finite, and year-to-year variability enters only as a uniform multiplier on the forcing.

The model resembles seasonally forced epidemic models, in which epidemics deplete susceptible people, births replenish them, and changes in the birth rate shift measles between annual, biennial, and irregular epidemics [6]; the resemblance is structural only.

The MATLAB code and LaTeX source for this report are available at https://github.com/lkaising/mars-dust-project.

5 Conclusion

In this spatially lumped model, a dusty season every year does not produce a global event every year. Each global event depletes the available surface dust, and modest year-to-year forcing differences make the intervals between events irregular. For the parameter values tested, these processes can produce intermittent large dust events.

References

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